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A REVOLUTION IN MOBILITY

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► **SEVERAL YEARS AGO**, I was at the immigration counter at Heathrow, London, and the officer wanted to know the purpose of my visit. I said I was there to attend a conference on mobility called “MOVE”. I had to clarify to the confused officer—it’s about driverless cars, air taxis, and the like. He’s not alone. People of an older generation are used to looking at cars and planes as belonging to the realm of transportation and would naturally be puzzled to hear the term “mobility”.

Much later, I was at a panel discussion when a co-panellist clarified the distinction between, “transportation” and “mobility”. Transportation is mode-centric, that is, what a bus service in a city would offer its riders. Mobility, on the other hand, is user-centric, regardless of the modes of transportation.

Mobility, then, has to function at different levels. The world is fast moving towards on-demand mobility (ODM), involving shared mobility, away from private ownership of vehicles. It is in this context that we have to envisage what the scenario on mobility would look like by 2047 in India.

While the known modes of transportation could get only better in the next two decades, the emphasis of this write-up is on modes that aren’t too prevalent today—micro-mobility, aerial mobility, and high-speed mobility. We will consider these below in the reverse order. Before that, we should point out two salient features of them: sustainability and efficiency. As with existing vehicles that will evolve towards electric (and possibly hydrogen) and become greener, the new forms too will decidedly be more carbon conscious. Besides, they will offer fast, hassle-free, and affordable on-demand throughput.

India is making epochal strides in the development of hyperloop technology. As we speak, we appear to be second only to China in the level of technology development on this front. From a present-day 450-m subscale vacuum tube-track with a magnetically propelled and levitated pod run nearing completion at IIT Madras’ Discovery Campus outside Chennai, we aim to leap to a full-scale demonstration up to the commercially viable speed of 600 kmph over approximately a 10 km distance within the next five years. The Indian Railways has